

Transparent [SimTalk] - Comment

Makes the background of the object designated by <Path> transparent in the *Frame* (**true**) or not transparent (**false**).

Type

Attribute

Syntax

```
<Path>.Transparent: boolean
```

Assignment Value

You can assign a value of data type *boolean*.

Specify **true** to show the *Comment* in the color that you selected as the **Font Color** on the background of the *Frame*.

Specify **false** to show the space around the text in white.

Example

```
MyComment.Transparent := false
```

See also

[Transparent \[check box\] - Comment](#)

[Font Color \[Comment\]](#)

Display

Display [object]

Use the object *Display* for presenting current data and results of the simulation run. The *Display* shows a value, for example of an attribute, at all times throughout the entire simulation run.

Image



Description

If the *Display* is not **active**, *Plant Simulation* shows its icon in the *Frame*, into which you inserted it. If it is active, the *Display* shows a value. In **Sample Mode** it updates its display periodically, in **Watch** mode it update it when the displayed value changes.

The *Display* shows the value either as text (string), as a bar or as a slice of a pie. Bar and pie display are only supported for numerical values. They are easy to comprehend, but not very well suited if accuracy is important.

For bar or pie representation the *Display* shows the value in relation to a specified range. An empty bar/pie indicates values less-than or equal to the lower bound of the range. A full bar/pie represents values beyond or equal to the upper bound of the range. Modify the overall size of the bar/pie representation by dragging its outline with the left mouse button while holding both the **Shift** and the **Ctrl** keys down.

The *Display* shows the actual minimal and maximal values as dashed lines which is not very accurate. In **Bar** and **Pie** mode it shows the current values in the text boxes **Minimum** and **Maximum** on the tab **Data**.

You can also show the numerical value of a *Display* in an **HtmlReport**, compare [Display an Object of Type Display](#).

Note

The *Display* only shows values if **MUs and States [Home ribbon]** is activated .

If **MUs and States** is deactivated, *Plant Simulation* prevents updating a number of values during the simulation, for example of objects of type *Variable* or *Display*.

Drag-and-Drop for Display Objects

Use drag-and-drop to accomplish these tasks:

To	Drag	To the
Show the statistics of a material flow object	the material flow object	<i>Display</i>
Show the current value of a <i>Variable</i>	the <i>Variable</i>	<i>Display</i>

To show a **tooltip** with information about the *Display*, hover with the mouse over it.

Fill Level Buffer 14

MyDisplay
Origin: .UserInterface.Display
Next event: UpdateDisplay (+45.7746)

To change the length of the graphic and the anchor points of the *Display*, click [Show Manipulators](#) on the **Edit** ribbon tab or press **M** on the keyboard.



Add the Object to the Simulation Model

To add the object *Display* to your simulation model, click  [Manage Class Library](#) > **Basic Objects** > **UserInterface** > **Display** on the **Home** ribbon tab.

See also

Show Values During the Simulation with the Display

Dialog Box of the Display

Dialog Box of the Display

Double-click the icon of the *Display* to open its dialog box.

Edit Simulation Properties

In the dialog box you can change the **simulation properties** of the object. The shared properties are described under [Dialog Items of the Objects](#).

Edit Animation Properties

To edit the **3D properties** of the object in the dialog box [Edit 3D Properties](#):

- Click the button  **Edit 3D Properties** in the lower left corner of the simulation properties dialog box.
- Select the object in the model and press the **spacebar**.

To manipulate the graphic of the object, click [Show Manipulators](#) on the **Edit** ribbon tab or press **M** on the keyboard.

See also

[Active \[check box\] - Display](#)

Active [check box] - Display

To activate the display of the *Display*, select this check box.

☒ Active

Remarks

If the *Display* is not active, *Plant Simulation* shows its **Name** in the *Frame* instead of a **Value**.

Active	Not Active
Fill Level Buffer 14	MyDisplay

To deactivate it, clear the check box or select **Deactivate** on the context menu.

Instead, you can also right-click the *Display* object in the *Frame* and select **Activate** on the context menu.

SimTalk

Active [SimTalk] - Display

Tab Data

Tab Data

The tab **Data** provides the settings listed in the table of contents to the left.

Path [text box] - Display

Click the ellipsis button and select the *attribute* or the *method* in the dialog **Select Object** whose value the *Display* displays.

Path ... 

Remarks

If you want to show the value of a global variable, you can type in the path to the *Variable* itself. You can also type in an expression, such as `buffer . NumMU+1`.

The *Display* supports the data types *boolean*, *integer*, *real*, *string*, *object*, *time*, *money*, *length*, *weight*, *speed*, *date*, and *dateTime*. If the path you type in is invalid, the background of the text box changes to red.

Note

F2 opens the dialog of the object, whose name you typed into the text box.

SimTalk

Path [SimTalk] - Display

See also

Select Object [for controls]

Comment [text box] - Display

Type a brief description of the displayed value into the text box **Comment**.

Comment 

Remarks

You might, for example, type in the name of the attribute that the *Display* displays, such as Fill Level Buffer.

The *Display* shows this text below the value in the *Frame* window.



Fill Level Buffer

SimTalk

Comment [SimTalk] - Display

Mode [drop-down list] - Display

Select the **Operating Mode** of the *Display*: **Sample** or **Watch**.

Mode	Watch Watch Sample	☐	Mode	Sample
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Remarks

You can select one of these settings:

- **Sample** mode updates the display with the frequency you type into the text box **Interval**.
- **Watch** mode updates the displayed value each and every time the value changes. If the path and the value you type in is not **watchable**, *Plant Simulation* does not update the display.

The column **Watchable** in the window **Show Attributes and Methods** shows which *attributes* and *read-only attributes* *Plant Simulation* can watch.

SimTalk

[Sampler \[SimTalk\]](#)

[SmpPeriod \[SimTalk\] - Display](#)

See also

[DisplayType \[SimTalk\]](#)

[update \[SimTalk\] - Display](#)

Interval [text box] - Display

Select **Sample Mode** to show the text box **Interval**. Type in the update frequency of the display into the text box. The format is the normal time format.

Mode	Sample	☐
Interval	1:00	☐

SimTalk

[SmpPeriod \[SimTalk\] - Display](#)

See also

[Mode \[drop-down list\] - Display](#)

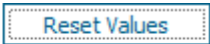
Value [Display]

Select **Bar** or **Pie** mode to show the **Current Value**.

Value 14

Minimum 0

Maximum 14

A rectangular button with a dashed border and the text "Reset Values" inside.

SimTalk

Value [SimTalk] - Display

Minimum [Display]

Select **Bar** or **Pie** mode to show the **Minimum** of the reached value.

Value 14

Minimum 0

Maximum 14

Reset Values

SimTalk

[GetMinimum \[SimTalk\]](#)

[GetMaximum \[SimTalk\]](#)

[resetMinMax \[SimTalk\]](#)

Maximum [Display]

Select **Bar** or **Pie** mode show the **Maximum** of the reached value.

Value	14	Minimum	0	Reset Values
		Maximum	14	

SimTalk

[GetMaximum \[SimTalk\]](#)

[GetMinimum \[SimTalk\]](#)

[resetMinMax \[SimTalk\]](#)

[Reset Values \[button\]](#)

Reset Values [button]

To reset the displayed minimum and maximum values shown on the tab, click this button.

Value	14	Minimum	0	Reset Values
		Maximum	14	

SimTalk

[resetMinMax \[SimTalk\]](#)

Tab Display

Tab Display

The tab **Display** provides the settings listed in the table of contents to the left.

Data		Display		User-defined	
Type	Text	Font Size	Extra Large		
Color	Text	Decimal Places	-1		
☐ Transparent	Bar				
	Pie				

Data		Display		User-defined	
Type	Text	Font Size	Extra Large		
Color		Decimal Places	-1		
☐ Transparent					

Type [drop-down list] - Display

Select how the *Display* displays the data it records.

Type Text
 Color Text
☐ Transpare Bar
Pie

Remarks

You can select one of these settings:

- **Text**

Shows the value as text. For floating point values you can type in the number of **decimal places**. The *Display* can show up to 15 decimal places. For the default setting **-1** it shows all existing places. Type in a positive value to show this number of places in the *Frame*.

Data **Display** User-defined

Type Text
 Color Text
☐ Transpare Bar
Pie

Font Size Extra Large
 Decimal Places -1

Type the text into the text box **Comment** on the tab **Data**.

Path Buffer.NumMU
 Comment Fill Level Buffer

- **Bar and Pie**

Maps the value in relation to the **Minimum** and the **Maximum** of the range that you specify. They visualize values inside the range by the height of a bar or a filled segment of the pie. An empty bar/pie corresponds to values below or equal to the lower bound. A full bar/pie represents values above or equal to the upper bound. To change the height and the width of the bar/pie graphic, hold down the **Shift** and die **Ctrl** keys and drag the border of the graphic with the left mouse button.

Plant Simulation converts the current value to the range, whose bounds you type in and shows it averaged.

Type in the **Maximum Value** of the displayed range of values.

Type in the **Minimum Value** of the displayed range of values.

Type: Bar
 Maximum: 50
 Minimum: 0

The *Display* shows the actual **Minimum** and **Maximum** values as dotted lines in the *Frame* window. This might not be as accurate as you need them to be. The *Display* shows the current values on the tab **Data** in the text boxes **Minimum** and **Maximum**.

Note

The *Display* can only show numerical values as a bar or a piece of a pie. These are easy to understand but not necessarily suited if accuracy is important.

To reset the values on the tab to 0, click **Reset Values** on the tab **Data**.

Text	Bar	Pie
Fill level of MyBuffer17 MyDisplay	 Fill Level Buffer	 Fill level Buffer

SimTalk

[DisplayType \[SimTalk\]](#)

[MinVal \[SimTalk\]](#)

[MaxVal \[SimTalk\]](#)

See also

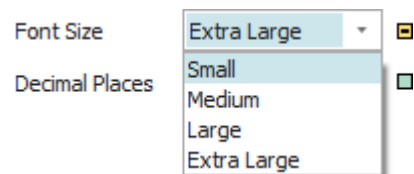
[GetMaximum \[SimTalk\]](#)

[GetMinimum \[SimTalk\]](#)

[resetMinMax \[SimTalk\]](#)

Font Size [drop-down list] - Display

Select the font size of the text which the *Display* displays for the **Type > Text**.



SimTalk

Font [SimTalk] - Display

See also

Type [drop-down list] - Display

Decimal Places [text box] - Display

If the **Type > Text** shows floating point values you can type in the number of decimal places.

The screenshot shows a configuration window with three tabs: 'Data', 'Display' (selected), and 'User-defined'. Under the 'Display' tab, there are four settings: 'Type' is a dropdown menu set to 'Text'; 'Color' is a color selection dropdown showing a blue color; 'Font Size' is a dropdown menu set to 'Extra Large'; and 'Decimal Places' is a text input field containing the value '-1'. Below these settings is a 'Transparent' checkbox which is currently unchecked.

Remarks

The *Display* can show up to 15 decimal places. For the default setting **-1** shows all existing places. Type in a positive value to show this number of places in the *Frame*.

SimTalk

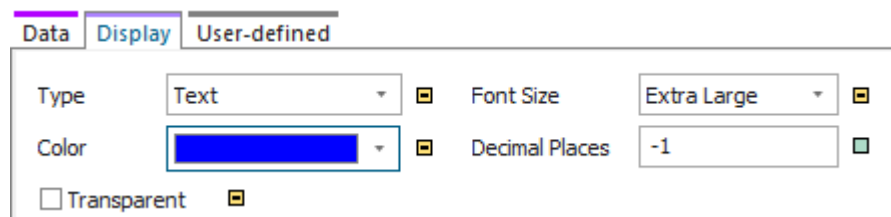
DecimalPlaces [SimTalk] - Display

See also

Type [drop-down list] - Display

Color [Display]

To select the color of the text, which the *Display* shows, and the color of the outline of its display graphic, click the drop-down arrow.



The screenshot shows a configuration window with three tabs: 'Data', 'Display', and 'User-defined'. The 'Display' tab is selected. It contains the following settings:

- Type:** Text (dropdown menu)
- Font Size:** Extra Large (dropdown menu)
- Color:** Blue (color selection dropdown)
- Decimal Places:** -1 (text input)
- Transparent:** ☐ (checkbox)

Remarks

You can select one of the predefined colors, or you can click **More Colors** and click the **Select** button to select a color in the color matrix. Then click **OK**. *Plant Simulation* shows this color next to **More Colors** and uses it as the active color.

SimTalk

Color [SimTalk] - Display

Transparent [check box] - Display

To make the background of a *Display*, which you inserted into a *Frame*, transparent, select this check box.

☒ Transparent 

Remarks

With a **Transparent Background**, the *Display* shows the space around the text in the color of the background image, or it shows the grid points of the *Frame*.

To show the space around the text in white, clear the check box.

Transparent	Not transparent
NextNumber= 28	NextNumber= 28

SimTalk

Transparent [SimTalk] - Display

Tab User-defined

Define your own attributes as described under the [Tab User-defined](#).

Navigate Menu

The commands are described under the [Navigate Menu](#).

View Menu

The commands are described under the [View Menu](#).

SimTalk

[updateDialog \[SimTalk\]](#)

Tools Menu

The **Tools Menu** provides commands to access its functions.

[Edit Controls](#)

[Edit Observers](#)

See also

[Tools Menu \[general description\]](#)

Help Menu

The commands are described under the [Help Menu](#).

Methods of the Display

_Methods of the Display

The *Display* provides:

- The *methods* listed in the table of contents to the left.
- The [Methods of All Objects](#).

To view all of the *methods*, *read-only attributes*, and *attributes* of the object, open the window [Show Attributes and Methods](#). The figure below illustrates the information using the example of the object *Station*.

Name	Value	Inherited	Watchable	Signature
contentsList	[]			([Contents:table]) -> any
createAttr				(NameOfUserDefinedAttribute:string, DataType:string) -> boolean
createIcon				(IconName:string, Width:integer, Height:integer) -> integer
deleteAttr				(NameOfUserDefinedAttribute:string) -> boolean

Name of the method
 Identifier and data type of the parameter you enter
 Data type of the return value

Name	Value	Inherited	Watchable	Signature
moveToFolder				(Folder:object) -> object
MU	VOID			([Index:integer:=1]) -> object
MUPart	VOID			([Index:integer:=1]) -> object

Name of the method
 Identifier and data type of the parameter you enter
 Default value
 Data type of the return value

- Select **Show Attributes and Methods** on the context menu of the **Class Library** to show the *methods*, *read-only attributes*, and *attributes* of the selected [Class \[general description\]](#).
- Press the **F8** key or click **Show Attributes and Methods** on the **Home** ribbon tab of the *Frame* into which you inserted an instance to show the *methods*, *read-only attributes*, and *attributes* of the selected [Instance \[general description\]](#).